Self-Check

8. When repairing a coworker's computer, you find some illegal files. What should you do next?
   a. Shut down the computer and unplug it.
   b. Take screenshots on the computer and save them in your own folder.
   c. Disconnect the computer from the network and leave it running.
   d. Delete the files.
9. Which backup site includes a running server that does not necessarily have access to the latest backups?
   a. Warm site
   b. Cold site
   c. Hot site
   d. On site
10. Which power device prevents a critical server from losing power, even for an instant?
   a. Surge protector
   b. Generator
   c. PDU
   d. UPS

Check your answers at the end of this chapter.

Summary

Section 12-1: Collect Network Data

- At its broadest, the term "network management" refers to the assessment, monitoring, and maintenance of all aspects of a network. It can include controlling user access to network resources, monitoring performance baselines, checking for hardware faults, ensuring optimized network performance for critical applications, maintaining records of network assets and software configurations, and determining what time of day is best for upgrading hardware and software.
- Sensors in each data room, equipment rack, or device chassis feed information to a physical device or software installed on a server that presents an administrative dashboard to network administrators.
- Some traffic monitoring tools will provide real-time analysis of data with alerts when conditions meet certain thresholds or when an anomaly is detected, while other tools are designed to log data for retroactive analysis only as needed. These historical traffic logs are primarily used to investigate network performance issues. The challenge with both real-time network monitoring and traffic logging is gaining access to the traffic itself.
- Cloud administrators endeavor to increase insights into their cloud infrastructure by taking an investigative approach to monitoring. Observability refers to the ability to capture internal system information in order to analyze and respond to it. Cloud observability must access data collected from a wide variety of monitoring services, many of which are cloud native in design.

Section 12-2: Manage Network Traffic

- To know when there's a problem on the network, you must first know what is normal for that network. A baseline is a report of the network's normal state of operation and might include a range of acceptable measurements.
- As a subset of performance management, bandwidth management refers to a collection of strategies to optimize the volume of traffic a network can support. These techniques might include any of the following technologies: flow control, congestion control, and QoS (quality of service).
- Rate-based flow control limits the amount of data that can be transmitted but does not provide feedback to the sender when that rate is exceeded. Instead, traffic is lost. In contrast, feedback-based flow control at the data link layer gives the sender some kind of indication when the transmission rate is exceeding the receiver's ability to handle the incoming traffic.
- Congestion control techniques are designed to prevent congestion before it occurs (called open-loop congestion control) and also to remedy congestion after it starts (called closed-loop congestion control).

- Network administrators must efficiently manage a network's QoS (quality of service) configurations, which is a group of techniques for adjusting the priority a network assigns to various types of transmissions. To do this, network administrators need to be aware of the applications used on a network, including the application protocols they use and the amount of bandwidth they require.

Section 12-3: Plan Response and Recovery Strategies

- An incident response plan specifically defines the characteristics of an event that qualifies as a formal incident and the steps that should be followed as a result. Incident response actually begins *before* the incident occurs.
- During some incidents, data will need to be collected in such a way that it can be presented in a court of law for the purpose of prosecuting an instigator of illegal activity. Some of the forensic data available for analysis can be damaged or destroyed if improperly handled. It's critical that every IT technician in a company knows how to safeguard sensitive information, logged data, and other legal evidence until the first responder or incident response team can take over the collection of evidence.
- When bad things happen, businesses need a plan to ensure business continuity, which is the ability of the company to continue doing business with the least amount of interruption possible. At a high level, a BCP (business continuity plan) defines the resources and protocols the business will use to continue providing service to its customers with little or no disruption during a disaster.
- Disaster recovery contingencies are commonly divided into these categories: cold site, warm site, hot site, and (more recently) cloud site. These options vary by the amount of hardware, software, planning, investment, and employee involvement each requires. They also vary according to how quickly they will restore network functionality in case a disaster occurs.
- Part of managing a network's availability involves managing the facilities and infrastructure that support the network, such as power connections and power sources during outages or fluctuations. Some of the devices involved in providing power to a network are focused on controlling and distributing power, while other devices store or generate power.
- A backup is a copy of data or program files created for archiving or safekeeping. Maintaining good backups is essential for providing fault tolerance and reliability. In deciding what to back up, you might be required by certain compliance standards to back up certain types of data for a specified period.

Key Terms

For definitions of key terms, check the Glossary.

3-2-1-1 Rule
alert
audit report
audit trail
backup
bandwidth management
baseline
BCP (business continuity plan)
buffering
business continuity
chain of custody
cloud site
cold site
community string
congestion control
contention
CoS (Class of Service)
delay-sensitive
DiffServ (Differentiated Services)
disaster
DR (disaster recovery) plan
DRaaS (Disaster Recovery as a Service)
DSCP (Differentiated Services Code Point)
first responder
flow control
flow data
hot site
incident
incident response plan
log
loss-tolerant
MIB (Management Information Base)
NetFlow
network management
NMS (network management system) server
observability
OID (object identifier)
packet analysis
PDU (power distribution unit)
power load
QoS (quality of service)
RPO (recovery point objective)
RTO (recovery time objective)
severity level
simulation
surge protector
system log
tabletop exercise
traffic analysis
traffic log
traffic policing
traffic shaping
trap
UPS (uninterruptible power supply)
validation test
warm site

Acronyms Checklist

The acronyms in Table 12-1 are listed in the CompTIA Network+ objectives and could appear on the Network+ exam. This means that exam questions might use any of these acronyms in context so that you must know the meaning of the acronym in order to answer the question correctly. Make sure you're familiar with what each acronym stands for and the general concept of the term itself. All these acronyms are used in context in this chapter.

Table 12-1 Chapter 12 acronyms

Acronym	Spelled out
DR	disaster recovery
MIB	Management Information Base
PDU	power distribution unit
QoS	quality of service
RPO	recovery point objective
RTO	recovery time objective
UPS	uninterruptible power supply

Review Questions

1. What command requests the next record in an SNMP log?
   a. SNMP Get Request
   b. SNMP Get Next
   c. SNMP Trap
   d. SNMP Get Response
2. What port do SNMP agents listen on?
   a. Port 161
   b. Port 21
   c. Port 162
   d. Port 20
3. What field in an IPv4 packet is altered to prioritize video streaming traffic over web surfing traffic?
   a. Traffic Class
   b. Priority Code Point
   c. Time to Live
   d. DiffServ
4. Which power backup method will continually provide power to a server if the power goes out during a thunderstorm?
   a. Online UPS
   b. Generator
   c. Dual power supplies
   d. Standby UPS
5. Which type of disaster recovery site contains all the equipment you would need to get up and running again after a disaster, and yet would require several weeks to implement?
   a. Warm site
   b. Standby site
   c. Hot site
   d. Cold site
6. Which log type is used to prove who did what and when?
   a. Traffic log
   b. Audit log
   c. System log
   d. Syslog
7. Which data link layer flow control method offers the most efficient frame transmission when sending large volumes of data?
   a. Go-Back-N sliding window
   b. Choke packet
   c. Selective Repeat sliding window
   d. Stop-and-Wait
8. What type of cloud service captures API calls made to change resource configurations?
   a. Logging service
   b. Alerting service
   c. Data analysis service
   d. Notification service
9. Which congestion control technique helps to prevent network congestion?
   a. Backpressure
   b. Explicit signaling
   c. Admission policy
   d. Implicit signaling
10. Which backup type, if performed daily, would offer the lowest RTO and why?
   a. Incremental backup
   b. Full backup
   c. Archive backup
   d. Differential backup

Scenario-Based Questions

Question 12-1

While troubleshooting a recurring problem on your network, you want to examine the TCP messages being exchanged between a server and a client. Which tool should you use on the server?

a. Spiceworks
b. Wireshark
c. iPerf
d. NetFlow

Question 12-2

One of your coworkers downloaded several very large video files for a special project they're working on for a new client. When you run your network monitor later this afternoon, what list will your coworker's computer likely show up on?

a. Top talkers
b. Top listeners
c. Giants
d. Jabbers

Question 12-3

Your roommate has been hogging the bandwidth on your router lately. What feature should you configure on the router to limit the amount of bandwidth their computer can utilize at any one time?

a. Power management
b. Congestion control
c. Flow control
d. Traffic shaping

Question 12-4

When you arrive at work one morning, your inbox is full of messages complaining of a network slowdown. You collect a capture from your network monitor. What documentation can help you determine what has changed?

a. SLA
b. DR plan
c. Baseline
d. Audit report

Hands-On Projects

Note 12-11 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Project 12-1: Configure SPAN and Syslog in Packet Tracer

Estimated time: 45 minutes
Objective 3.2: Given a scenario, use network monitoring technologies.

Resources:

- Cisco Packet Tracer

Description:

In this chapter, you read about various ways to capture network traffic for analysis and monitoring. In this project, you'll experiment with two of these technologies: SPAN on switches and a syslog server. This Packet Tracer project starts with a new network and is *not* intended to build on your Packet Tracer network for the Capstone Projects. For this project, do *not* use your existing Packet Tracer network (most recently worked on in Chapter 10). In this project, start with a new Packet Tracer file to experiment with SPAN and syslog. You will *not* need to save this network for future projects or chapters.

Complete the following steps:

1. Open Cisco Packet Tracer—make sure you're starting from a new file, not an existing network.
2. In your Packet Tracer workspace, add a 2960 switch and three PCs.
3. Configure static IP addresses on all three PCs within the same subnet. What IP addresses did you assign each PC?
4. Connect the PCs to the first three switch ports (FastEthernet 0/1 - 0/3). Wait for all connections to come up.
5. In the bottom-right corner, click **Simulation**. This opens the Simulation Panel.
6. By default, the simulation will display all messages from all protocols on the network once you start the simulation—you can check the list of applicable filters in the "Event List Filters - Visible Events" section. For this project, you only want to display ICMP messages. At the bottom of the Simulation Panel, click **Show All/None**, which clears all visible event types. Click **Edit Filters**. In the Cisco Packet Tracer dialog box, check the box on the IPv4 tab for **ICMP**. Close the Cisco Packet Tracer dialog box. Confirm ICMP is the only visible event type listed.
7. When you start the simulation in the next step, you will run pings between PCs on your network, and the results will be displayed in the PDU List Window in the bottom-right corner of your Packet Tracer interface. For a more convenient arrangement, in the bottom-middle pane, click **Toggle PDU List Window** to move this pane to a larger space in your Packet Tracer interface. If desired, you can also grab the top bar of the Simulation Panel to move this module around on your screen in a separate window.
8. On the common tools bar, click the **Add Simple PDU (P)** button, which looks like a closed envelope. This will generate a ping-based conversation between two devices without having to enter commands at a CLI. Click **PC0** as the source device and click **PC1** as the destination device.
9. In the Simulation Panel, click the **Play (Alt + P)** button. As the simulation begins, a PDU leaves PC0 and arrives at the switch. Observe the traffic carefully. To which device does the switch send the first PDU? Why do you think this is?
10. At the bottom of the interface, click the **Delete** button to stop the simulation for this scenario. Click **Realtime** to return to Realtime mode.

Now you're ready to add a sniffer to the network that will monitor all traffic on the switch. Complete the following steps:

11. From the End Devices group, add a **Sniffer** to the workspace.
12. Connect the sniffer's Ethernet0 port to the switch's FastEthernet 0/24 port. The sniffer does not need an IP address to do its job.

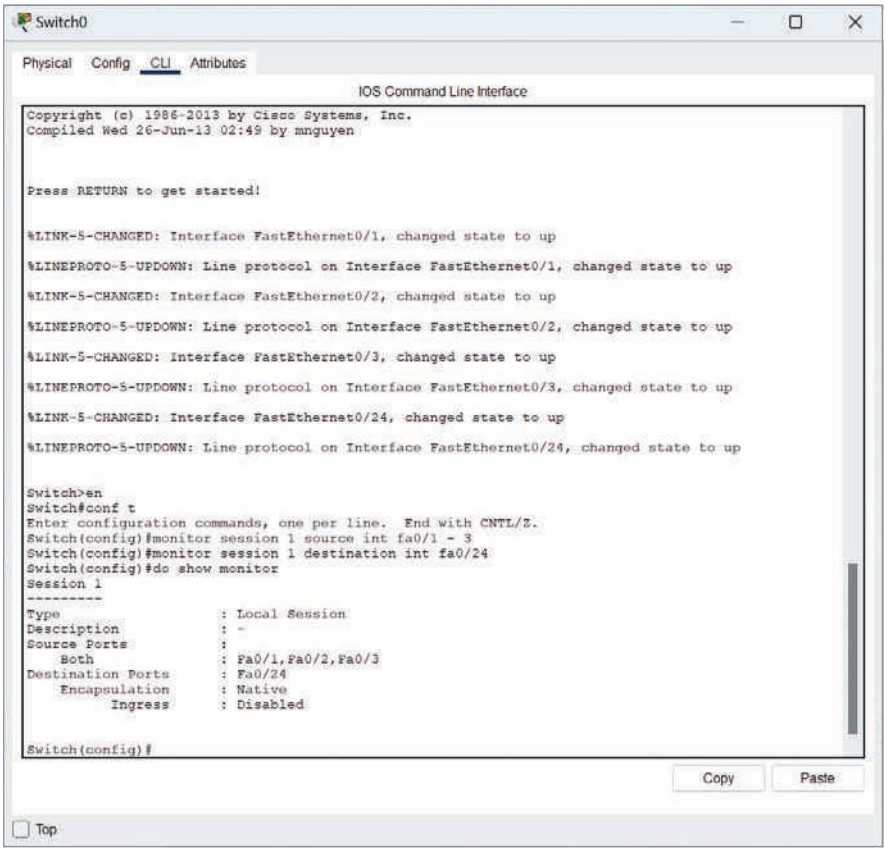
With these devices connected to your network, you're ready to configure a SPAN monitoring session on the switch. Complete the following steps:

13. On the switch's **CLI** tab, enter the commands in Table 12-2. Depending on the switch's status, you might first need to press **Enter** to access the command prompt.

Table 12-2 Configure a SPAN monitoring session on a switch

Command	Purpose
`enable`	Enables privileged EXEC mode
`configure terminal`	Enters global configuration mode
`monitor session 1 source int fa0/1 - 3`	Configures source interfaces for the monitoring session
`monitor session 1 destination int fa0/24`	Configures the destination interface for the monitoring session (there can be only one)
`do show monitor`	Displays the monitoring session configuration; confirm your configuration matches Figure 12-35

Figure 12-35 SPAN monitoring session from three interfaces to FastEthernet0/24



Source: Cisco Systems, Inc.

14. To test your monitoring session, enter Simulation Mode and send a simple PDU from PC0 to PC1 again. Observe the traffic carefully. To which device(s) does the switch send the PDU this time? Why do you think this is?
15. Pause the simulation after three or four messages have been sent across the network.
16. To display the captured traffic, click the **Sniffer** and click its **GUI** tab. Apply the same filter here as the one you used for Simulation Mode in Step 6 and run the ping again. Pause the simulation after three or four messages have been sent across the network.
17. On the sniffer's GUI tab, scroll down in the left pane, and click any of the captured ICMP messages listed at the end of the list. What are the source and destination IP addresses of the message you chose?

Syslog is another way to capture network traffic; however, syslog functions differently by capturing messages processed on a monitored device. To compare SPAN and syslog functionality, complete the following steps:

18. Return to Realtime mode. Add a 2901 router and a server to the workspace. Connect both to the switch.
19. On the server's **Services** tab, confirm the Syslog service is turned on. Configure the server with a static IP address on the same subnet as your PCs. What IP address did you give the server?
20. Configure the router's interface connected to the switch with a static IP address in the same subnet as the other devices, and turn it on.

Syslog logging is enabled by default on the router, but additional information is needed to send the required logs to the server. Complete the following steps:

21. To configure syslog logging on the router's CLI, enter the commands listed in Table 12-3, starting in global configuration mode.

Table 12-3 Configure syslog on a router

Command	Purpose
`logging <server IP address>`	Tells the router where to send syslog messages
`logging trap debugging`	Requires all messages from the debugging level (level 7, which is the lowest severity level) and above to be logged with the server; note that Packet Tracer only supports logging at level 7 (debugging), which includes all severity levels
`exit`	Returns to privileged EXEC mode
`debug ip icmp`	Enables debugging for ICMP messages
`show debugging`	Displays debugging configuration; confirm ICMP packet debugging is on
`show logging`	Displays syslog configuration

22. Check the logging configuration on your router. Where are the router's logs being sent? What port are these messages using?
23. To test your syslog configuration, ping one of your PCs from your router. Return to your syslog server's Services tab on the Syslog page to examine captured traffic. **Take a screenshot** of the Syslog Service window showing logged traffic; submit this visual with your responses to this project's questions. How many ICMP messages are logged?

24. Syslog captured the ICMP messages for a ping between the router and a PC. Now ping between two PCs. Return to your syslog server to examine captured traffic. How many ICMP messages are logged now? Why do you think this is?
25. Save this Packet Tracer file in a safe place for future reference. Make some notes on your Wikidot website about your activities in Packet Tracer for this project.

Note to instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors site.

Project 12-2: Organize Your Wikidot Website

Estimated time: 45 minutes (+15 minutes for group work, if assigned)
Objective 3.1: Explain the purpose of organizational processes and procedures.
Group work: This project includes enhancements when assigned as a group project.
Resources:
- Internet access

Description:
Throughout this course, you've kept notes on various projects in your Wikidot website. To finish these projects, let's do some final cleanup and organization so these notes will continue to be useful and easily accessible for you as you move into your other IT classes and your career. You might decide to continue adding notes, pages, and categories, or create new wikis as needed. Using good organization and adding thorough notes could make your wiki a valuable exhibit when applying for your first job in IT. Complete the following steps:

1. Sign into your account at **wikidot.com**. Return to your wiki site.
2. First, adjust the side navigation menu. On the All Pages page (click the gear icon and click **List All Pages**), click **Side Navigation** and then click **Edit**. Your text should match the text shown in Figure 12-36. Throughout this project, you can ignore any red, squiggly lines in the text unless they indicate a place where you've misspelled a word. Be sure to use correct spelling and good grammar in your wiki's content text.

Figure 12-36 Side navigation bar coding

```
[[div class="text-center" style="margin-bottom: 12px;"]]
[[module Clone]]
[[/div]]

+ Site Navigation

* [[[help:_home|User Guide]]]
* [[[help:first-time-user|First Time User]]]

* [[[main:about|About]]]
* [[[main:contact|Contact]]]
* [[[legal:_home|Legal]]]
* [[[forum:start|Discussion Forums]]]
* [[[system:members|Members]]]

+ Create a Page
[[module NewPage size="20" button="New page"]]
```

Source: Wikidot Inc.

3. Change the User Guide link to point to the list of all pages instead. On the User Guide line (the first line under "+ Site Navigation"), edit it to read as follows:

```
* [[[system:list-all-pages|All Pages]]]
```

4. When you're finished, the edited text should match the first item in the Site Navigation list in Figure 12-37. Save your changes and test the new link under Site Navigation on the right.

Figure 12-37 Edits for side navigation bar

```
[[div class="text-center" style="margin-bottom: 12px;"]]
[[module Clone]]
[[/div]]

+ Site Navigation

* [[[system:list-all-pages|All Pages]]]
* [[[help:first-time-user|First Time User]]]

* [[[main:about|About]]]
* [[[main:contact|Contact]]]
* [[[legal:_home|Legal]]]
* [[[forum:start|Discussion Forums]]]
* [[[system:members|Members]]]

+ Create a Page
[[module NewPage size="20" button="New page"]]
```

Edited line

Source: Wikidot Inc.

Each time you named a page with one part before the colon and another part after the colon, such as Applications:Wireshark, you added that page to a category. The first name, such as Applications, is the name of the category. The second name, such as Wireshark, is the name of the page. Complete the next step to display all available categories:

5. To see a list of all categories in your wiki, click the gear icon and then click **Site Manager**, **Appearance & Behaviour**, and **Navigation elements**. Click the arrow next to "Choose the category."

Here, the available categories are listed, including the ones you've created. The trick now is to find a way to list pages according to each category. To do this, you'll first need to create a Categories page. Complete the following steps:

6. Go back to your wiki and create a new page called **system:All Categories**.
7. Add the text `[[module Categories]]` to your page and save it.
8. Go back to the **All Pages** list. The All Categories page is listed as "system:All Categories." To change the page name to "All Categories" instead, click the **system:All Categories** page, click **Edit**, and change the title to **All Categories**, and save your changes. The *name* of the page still includes its category ("system"), but now the *title* of the page is simply "All Categories."

The Categories module automatically creates a list of all the categories and all the pages within each category. If any page is listed in the wrong category, you can't edit the page's title to change its category. When you edit the page, you're editing the title of that page, not its name, which is what defines the page's category. Instead, use the following step to edit a page's category:

9. Click the plus (**+**) symbol to display the pages for each category you created, including "applications" and "virtualization," to confirm all pages you created are listed in the correct categories.
10. If a page is listed in the wrong category, go to the page and click **+ Options** and then click **Rename**. Change the page's category, which is the name *before* the colon, and click **Rename/move**. Repeat for any other miscategorized pages.

Now let's edit the top navigation bar so it shows one or more categories as an option, and each page within that category as an option. Complete the following steps:

11. Click the gear icon and click **Edit Top Bar**.
12. Click the **Edit** button on this page. Your text should be similar to the text shown in Figure 12-38.

Figure 12-38 Top navigation bar coding

```
[[ul class="nav navbar-nav"]]
[[li]][[a href="/main:about"]]About [[span class="fa fa-info-circle"]]@< >@[[/span]][[/a]][[/li]]
[[li]][[a href="/main:layout"]]Layout [[span class="fa fa-code"]]@< >@[[/span]][[/a]][[/li]]
[[li]][[a href="/system:join"]]Membership [[span class="fa fa-user"]]@< >@[[/span]][[/a]][[/li]]
[[li]][[a href="/help:_home"]]User Guide [[span class="fa fa-exclamation-circle"]]@< >@[[/span]]
[[/a]][[/li]]
[[li class="dropdown"]]
[[a href="#" class="dropdown-toggle" data-toggle="dropdown"]]Help Docs [[span class="fa fa-question-
circle"]]@< >@[[/span]][[/a]][[ul class="dropdown-menu"]]
  [[li]][[a href="/help:_home"]]User Guide[[/a]][[/li]]
  [[li]][[a href="/help:first-time-user"]]First Time User[[/a]][[/li]]
  [[li]][[a href="/help:quick-reference"]]Quick Reference[[/a]][[/li]]
  [[li]][[a href="/help:creating-pages"]]Creating Pages[[/a]][[/li]]
  [[li]][[a href="/help:editing-pages"]]Editing Pages[[/a]][[/li]]
  [[li]][[a href="/help:navigation-bars"]]Navigation Bars[[/a]][[/li]]
  [[li]][[a href="/help:using-modules"]]Using Modules[[/a]][[/li]]
  [[li]][[a href="/help:templates"]]Templates[[/a]][[/li]]
  [[li]][[a href="/help:css-themes"]]CSS Themes[[/a]][[/li]]
[[/ul]]
[[/li]]
```

Source: Wikidot Inc.

Currently, the only link in the top navigation bar that gives a drop-down menu is the Help Docs link. In the next few steps, you'll remove some of the links in the top navigation bar and add a drop-down link for each category. Complete the following steps:

13. Delete the **Layout** and **Membership** lines in this text (lines 3 and 4).
14. On the next line, change the text **User Guide** to **All Categories**. Change its location to **`system:all-categories`**. The line should now read as follows:

    `[[li]][[a href="/system:all-categories"]]All Categories [[span class="fa fa-exclamation-Circle"]]@< >@[[/span]][[/a]][[/li]]`
15. On the dropdown-toggle line, change the text **Help Docs** to the name of one of your categories, such as **Applications**.

For this step, you'll create drop-down items for the drop-down link you just created. It might help to have two browser windows open—one showing the All Pages list for a reference and the other showing the top navigation menu editing page. Add extra lines if needed and remove any sub-item lines you don't use. Complete the next step:

16. For each sub-item, add the name and location of a page within that category. For example, the Wireshark page would be listed under Applications like this:

    `[[li]][[a href="/applications:wireshark"]]Wireshark[[/a]][[/li]]`

    The Packet Tracer line will read like this:

    `[[li]][[a href="/applications:packettracer"]]Packet Tracer[[/a]][[/li]]`

Notice the small icons next to each item on the top navigation menu, such as an "i" in a circle, an exclamation mark in a circle, and a question mark in a circle. To change some of these icons, complete the following steps:

17. On the Applications line (the line that includes the dropdown-toggle text), change the text that reads `fa fa-question-circle` so it reads `fa fa-info-circle`.
18. On the About line (line 2), change the text that reads `fa fa-info-circle` so it reads **`fa fa-question-circle`**.
19. When you've made all these changes, review the navigation menu again and correct any typos or missed links. When you're ready, click **Save**. The new Top Navigation bar shows in the page's content area and at the top of the page.

20. Test each link to make sure it works correctly, and troubleshoot any problematic links. To make changes, go back to the **Top Navigation** page and click **Edit**. The edited text should be similar to Figure 12-39. Make sure the page addresses are typed exactly right.

Figure 12-39 The edited top navigation bar with one category updated

```
[[ul class="nav navbar-nav"]]
[[li]][[a href="/main:about"]]About [[span class="fa fa-question-circle"]]@< >@[[/span]][[/a]]
[[/li]]
[[li]][[a href="/system:all-categories"]]All Categories [[span class="fa fa-exclamation-
circle"]]@< >@[[/span]][[/a]][[/li]]
[[li class="dropdown"]]
[[a href="#" class="dropdown-toggle" data-toggle="dropdown"]]Applications [[span class="fa fa-info-
circle"]]@< >@[[/span]][[/a]][[ul class="dropdown-menu"]]
  [[li]][[a href="/applications:wireshark"]]Wireshark[[/a]][[/li]]
  [[li]][[a href="/applications:packettracer"]]Packet Tracer[[/a]][[/li]]
[[/ul]]
[[/li]]
[[li]][[a href="/main:contact"]]Contact [[span class="fa fa-envelope"]]@< >@[[/span]][[/a]][[/li]]
[[li class="dropdown admins-only"]]
[[a href="#" class="dropdown-toggle" data-toggle="dropdown"]][[span class="fa fa-
cog"]]@< >@[[/span]][[/a]][[ul class="dropdown-menu"]]
  [[li]][[a href="/_admin"]]Site Manager[[/a]][[/li]]
  [[li]][[a href="/nav:top"]]Edit Top Bar[[/a]][[/li]]
  [[li]][[a href="/nav:side"]]Edit Side Bar[[/a]][[/li]]
  [[li]][[a href="/css:_home"]]CSS Manager[[/a]][[/li]]
```

Source: Wikidot Inc.

21. Add more categories and page links, as desired, until you've listed all your categories and pages that you created for projects in this text.

22. On the top navigation bar, add a link to the All Pages page, with an information circle next to it. This link will make the All Pages page accessible directly from the Home page and most other pages in the wiki. What line of code must you add to the top navigation bar's code to accomplish this task?

23. Edit the Home page text and the About page text to reflect what you've accomplished during this course and to describe the information available in your wiki. Make any other changes you would like to the navigation menus, categories, or pages. You might add screenshots or photos to some of the pages, add more detailed notes, or create new categories for other projects you've completed in this and other courses. If desired, research other editing options, themes, codes, and modules so this wiki reflects your interests and learning progress. Plan to share a link to your wiki with prospective employers to showcase your learning progress and work experience. With this goal in mind, make sure your wiki presents a professional perspective of who you are and what skills you offer.

24. **For group assignments:** Invite a team member to check out your wiki and test your links. Correct any problems they find. Also review your teammate's wiki and report any errors you find. Exchange notes and ideas for ways to improve your wikis. What suggestions did your teammate have for your site? What suggestions did you give your teammate for their site?

25. When you're ready, **take a screenshot** of your wiki showing the top navigation and side navigation panes; submit this visual with your responses to this project's questions. Providing a link to your wiki when applying for an IT job could make a strong, positive first impression on a potential employer!

Capstone Projects

Note 12-12 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Capstone Project 12-1: Use Syslog in Ubuntu Desktop

Estimated time: 30 minutes

Objective 3.2: Given a scenario, use network monitoring technologies.

Resources:

- Access to the same computer used to complete Capstone Project 2-1

Description:

In this project, you will display and manipulate log file entries on a computer running the Linux operating system. Because Linux versions vary in the type of GUI application that allows you to open the system log, this exercise uses the CLI instead. For this exercise, you need a computer with a Linux operating system installed, such as the Ubuntu Desktop VM that you created in Chapter 2, Capstone Project 2-1. It need not be connected to a network, but for best results, it should be a computer that has been used in the past and not a fresh install. You must be logged on to the Linux computer as a user with administrator privileges. Complete the following steps:

1. Start your Ubuntu Desktop VM and open Terminal.
2. The syslog file contains information similar to that shown in Figure 12-40. The first step in viewing your Linux computer's system log is to find out where the file is located. Try each of these commands until you find the syslog file that contains information similar to that shown in Figure 12-40:

   more /etc/syslog.conf

   more /etc/rsyslog.conf

   more /etc/rsyslog.d/50-default.conf

Figure 12-40 Log files and their locations

```
#
# First some standard log files.  Log by facility.
#
auth,authpriv.*                 /var/log/auth.log
*.*;auth,authpriv.none          -/var/log/syslog
#cron.*                         /var/log/cron.log
#daemon.*                       -/var/log/daemon.log
kern.*                          -/var/log/kern.log
#lpr.*                          -/var/log/lpr.log
mail.*                          -/var/log/mail.log
#user.*                         -/var/log/user.log
```

Source: Canonical Group Limited

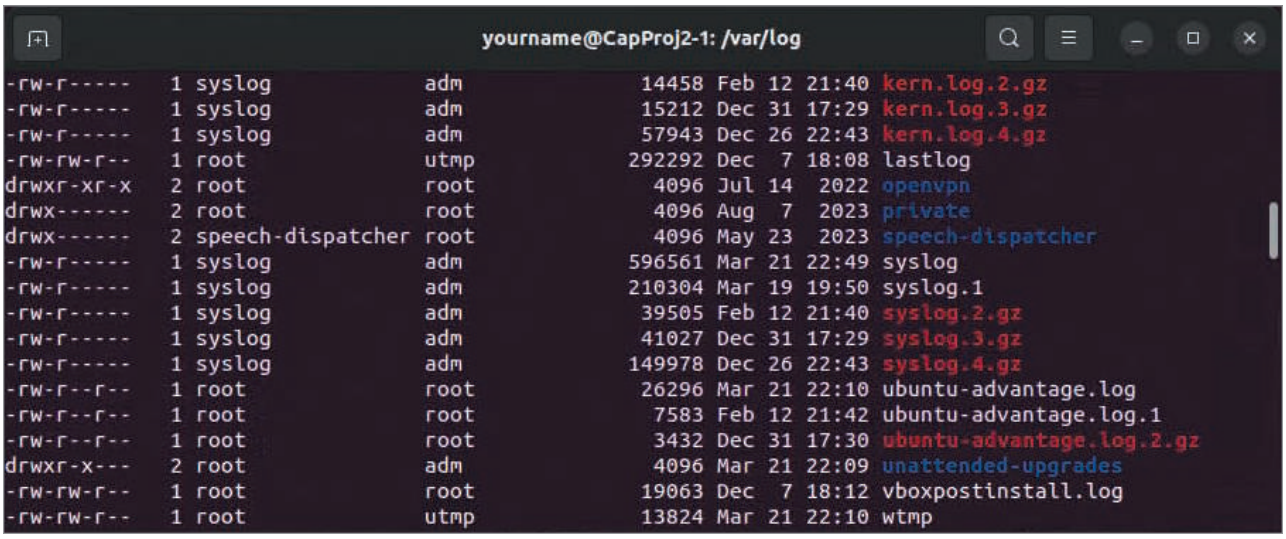
3. The first part of the syslog file appears. In this part of the file, there should be a list of log types and their locations, similar to the listing shown in Figure 12-40. If the listing is not visible in this part of the file, press the **Enter** or **Spacebar** key until it appears.
4. Write down the location and filename of the file that logs all events, as indicated by *.* in the first column. For example, it might be /var/log/syslog or /var/adm/messages.
5. Press the **Spacebar** enough times to view the entire log configuration file and return to the shell prompt.

6. Now that you know the name and location of your system log, you can view its messages. At the shell prompt, enter the command **`tail`** followed by the log's location. For example:
   - If your log file is at /var/log/syslog, enter **`tail /var/log/syslog`**
   - If your log file is at /var/adm/messages, enter **`tail /var/adm/messages`**

   If permission is denied, place **`sudo`** in front of the command, like this: **`sudo tail /var/log/syslog`**. You will then have to enter your user account password.
7. The last 10 lines of your log file appear (assuming it is at least 10 lines long). What types of messages are recorded? When did the events occur?
8. Find out all the types of log files your computer saves. Enter one of the following to change your working directory to the same directory where log files are kept:
   - If your log file is in the /var/log directory, enter **`cd /var/log`**
   - If your log file is in the /var/adm directory, enter **`cd /var/adm`**
9. To view a listing of the directory's contents, enter **`ls -la`**. What are three log files that appear in this directory?
10. Suppose you want to find every message in the system log file that pertains to DHCP addressing. At the shell prompt, enter one of the following:
    - If your log file is named syslog, enter **`grep DHCP syslog`**
    - If your log file is named messages, enter **`grep DHCP messages`**

    If permission is denied, place **`sudo`** in front of the command, like this: **`sudo grep DHCP syslog`**. A list of messages containing the term "DHCP" appears, if there are any.
11. Re-enter the **`tail`** command from Step 6, and then run a new **`grep`** search using a text string that appears in your results. What command did you use? **Take a screenshot** of your results showing successful location of the text string; submit this visual with your responses to this project's questions.
12. If your operating system is configured to start a new log file each day or each time the computer is restarted, your log file might be brief. Repeat Step 9, and this time, try to find other versions of the syslog or messages file in your working directory. For example, Ubuntu Linux will save older system messages in a file called syslog.1.gz, syslog.2.gz, and so on (Figure 12-41). If you find a larger, older log file, repeat Step 10 using this log file's name. How do the results differ?
13. Power off your Ubuntu VM. Make some notes on your Wikidot website about your activities for this project.

Figure 12-41 Several older syslog files are listed



Source: Canonical Group Limited

Capstone Project 12-2: Explore Observability Services in AWS

Estimated time: 45 minutes

Objective 3.2: Given a scenario, use network monitoring technologies.

Resources:

- AWS account created in Hands-on Project 7-1
- Internet access

Description:

In Chapter 7, you created an AWS account—either a standard account or an account through AWS Academy—and performed some tasks in the cloud. In this project, you will explore observability services in the AWS environment. Use the same AWS account you created in Chapter 7.

Caution | Depending on the status of your account and the selections you make during this project, some cloud resources can deplete your credits or accrue charges. Make sure to follow these steps carefully and delete all created resources at the end of this project, as directed.

Complete the following steps:

1. Sign into your AWS management console. If you're using an AWS Academy Learner Lab, you'll need to access your AWS management console through your AWS Academy course. If you're using a standard AWS account, sign in directly at **aws.amazon.com**.

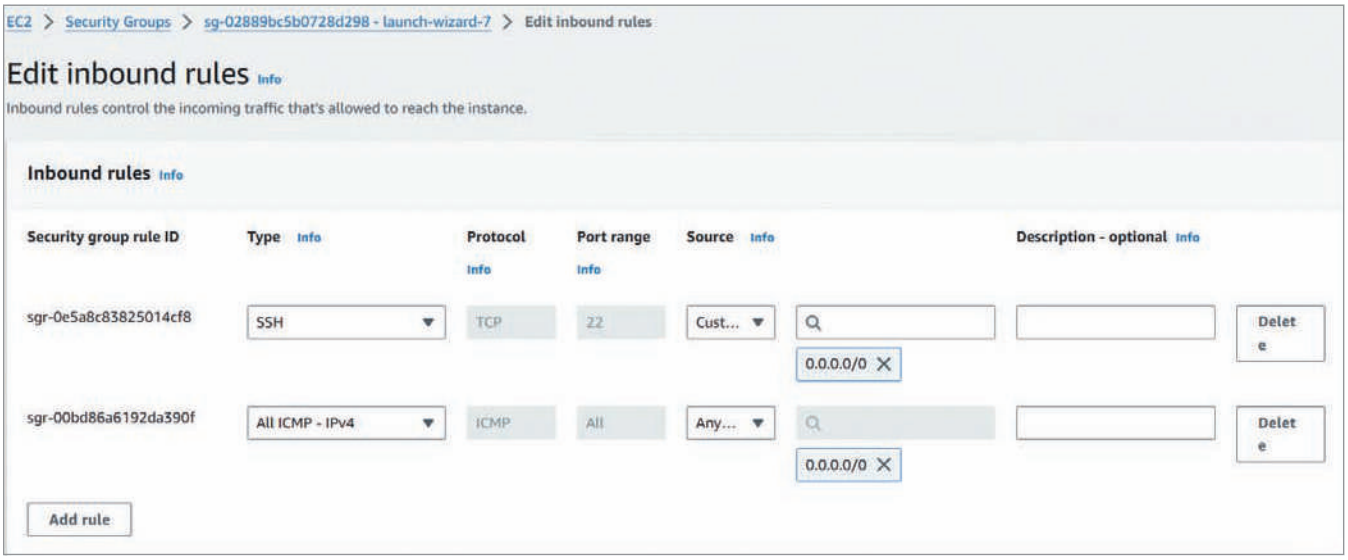
In AWS, many services are designed to help customers track their cloud resources, expenses, and activities. These services target specific purposes and are intended to work together for customers to gather all the observability information they need. Complete the following steps to begin exploring observability services in AWS:

2. The AWS Health Dashboard gives you insights into your account's health, such as relevant issues or scheduled changes. In the AWS console, in the Search box, enter **health dashboard** and click **AWS Health Dashboard**.
3. By default, you should be directed to the "Open and recent issues" page on the health dashboard. Most likely, there are no issues listed here. However, there might have been some events recorded. Click **Event log**. How many events are listed here?
4. Select one of the listed events, read about it, and then respond to the following questions:
   a. What activity prompted the discovery of the event?
   b. What was the scope of impact of this event?
   c. How long did the event last?

CloudWatch is one of the most important monitoring services in AWS. It can be used to analyze logs, automate responses to events, and trigger notifications to administrators. To create a CloudWatch alert on an EC2 instance's network utilization, complete the following steps:

5. Create an Amazon Linux EC2 instance using the t2.micro instance type; you do not need a key pair for this instance. For a refresher on how to create an EC2 instance, refer to Hands-on Project 7-2 in Chapter 7.
6. Once your instance is created, display the list of EC2 instances and click **View alarms** to confirm there are no existing alarms on your new instance.
7. To allow ICMP traffic to the instance, select the instance in the list of instances, and click its **Security** tab. Click the instance's security group.
8. On the security group's Inbound rules tab, click **Edit inbound rules**, and then click **Add rule**. Change "Custom TCP" to **All ICMP - IPv4**. In the Source field, select **Anywhere-IPv4** (Figure 12-42). Click **Save rules**.

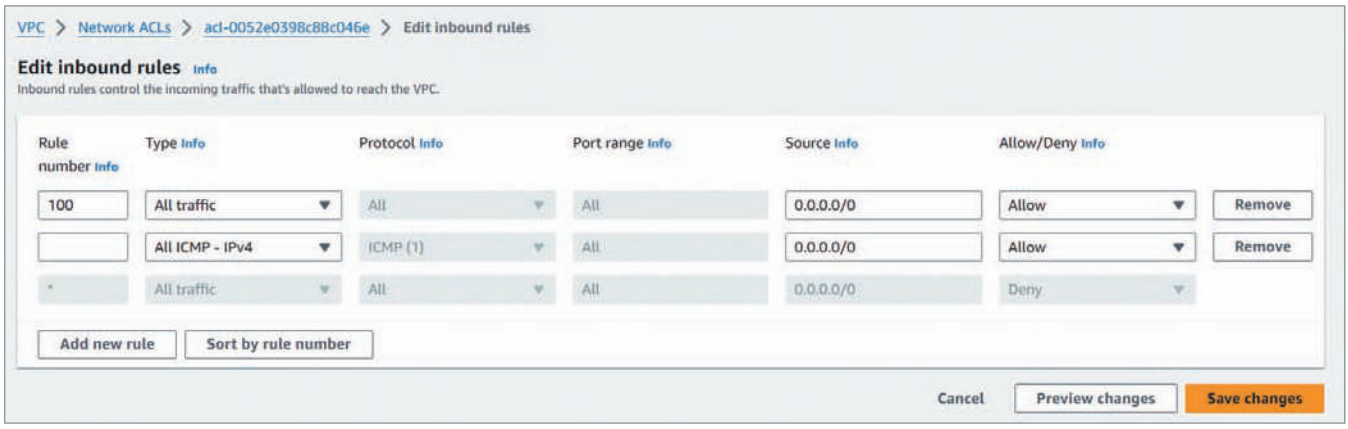
Figure 12-42 New security group rule to allow ICMP traffic



Source: Amazon Web Services, Inc.

9. You also need to allow ICMP packets through the VPC's NACL. In the AWS console, in the Search box, search for and click **VPC**.
10. In the navigation pane, click **Network ACLs**. Select the network ACL and then click its **Inbound rules** tab. Click **Edit inbound rules**, and then click **Add new rule**.
11. Type **90** in the Rule number field. Change "Custom TCP" to **All ICMP - IPv4** (Figure 12-43). Click **Save changes**.

Figure 12-43 New NACL rule to allow ICMP traffic



Source: Amazon Web Services, Inc.

12. Return to the EC2 dashboard. Select the instance in the list of instances, and click its **Monitoring** tab. What are three recommended alarms on this page?
13. In the AWS console, in the Search box, search for and click **CloudWatch**. In the navigation pane, click **All alarms**.
14. Click **Create alarm**, and then click **Select metric**. You'll be creating an alarm to monitor incoming network packets on your EC2 instance. Click **EC2**, and then click **Per-Instance Metrics**.
15. In the "Search for any metric, dimension, resource id or account id" search box, enter **network**, and select **NetworkPacketsIn**. Then click **Select metric**.

16. You need to configure the metric conditions for the alarm. In the Metric section, change the Period to **1 minute**. In the Conditions section, in the "Define the threshold value" field, enter **20**. Click **Next**.
17. In the Notification section, select "Create new topic." Give the topic a name, enter your email address in the field "Email endpoints that will receive the notification...", and then click **Create topic**.
18. You should receive an email almost immediately from AWS Notifications requesting subscription confirmation. Click the link in the email to confirm the subscription.
19. Scroll through the remainder of this page before moving forward. What are two automated response options listed here?
20. Click **Next**. Give the alarm a name, and then click **Next**. Then click **Create alarm**.
21. To trigger the alarm, you need to send several packets to your EC2 instance. On the instance's **Details** page, find its public IPv4 address. On your local computer, open a CLI and ping this address. Repeat the ping six or seven times within a minute to make sure that at least two dozen ping messages are sent to the instance.
22. When you receive the alarm notification, **take a screenshot** of the information provided in the email; submit this visual with your responses to this project's questions.
23. Before ending the lab, make sure to delete all resources created in this project, including the CloudWatch alarm and the EC2 instance. You do not need to delete the VPC, NACL, or security group.
24. If you used AWS Academy for this project, be sure to end the lab session in the AWS Academy console.

Caution | Depending on the status of your account and the selections you make during this project, some cloud resources can deplete your credits or accrue charges. Make sure to follow these steps carefully and delete all created resources at the end of this project, as directed.

Solutions to Self-Check Questions

Section 12-1: Collect Network Data

1. Which of the following would an environmental monitoring system track?
   Answer: a. Liquid detection
   Explanation: Monitoring sensors can **detect liquid** such as when water floods a room, identify if a room's lights are on, and monitor voltage from a UPS. However, an environmental monitoring system would not be responsible for tracking user authentication to the network, CPU utilization on individual devices, or network traffic characteristics (such as jabbers, runts, or giants).

2. Which log type would most likely be used first to investigate the cause of high numbers of dropped packets?
   Answer: a. Traffic log
   Explanation: Historical **traffic logs** can be used to investigate security breaches, confirm regulatory compliance, and investigate network performance issues. A system log tracks information on a single device and will not be helpful until you know which device(s) to check. The data in an audit log is thorough enough to retroactively prove compliance and is often used in forensics investigations; due to the level of detail included, an audit log is not likely the best place to start investigating a network problem. There's no such thing as a jitter log.

3. Which of the following would be assigned an OID?
   Answer: b. A switch's interface
   Explanation: A device, such as a web server, can be managed by an NMS server, typically using UDP ports 161 and 162. Each managed device may contain several managed objects, which can be any characteristic of the device that is monitored, including components such as a **switch's interface**, processor temperature, or memory utilization. Each managed object is assigned an OID (object identifier).

4. Which of the following cloud services would display a line chart indicating web server traffic over the previous 30 days?
   Answer: d. Dashboard
   Explanation: Real-time monitoring data and log data, such as historical web server traffic, can be streamed to **dashboards** for human evaluation. Notifications might be transmitted either by email or text message, also called SMS, or they can automatically prompt support ticket generation; notifications typically do not contain historical data, only data for the specific identified issue which triggered the alert. An alert is a message generated when a predefined event occurs, and these alerts might generate notifications to IT personnel or other automated responses. A log is a record of activities or state changes on a device or in a system; logs contain historical data that can be displayed in a line chart, but the logs themselves don't contain line charts.

Section 12-2: Manage Network Traffic

5. Which bandwidth management technique limits traffic specifically between a single sender and a single receiver?
   Answer: d. Flow control
   Explanation: **Flow control** addresses the number of frames that can be handled from end to end between a single sender and a single receiver. Congestion control addresses traffic throughout the network. QoS (quality of service) techniques allow for more nuanced control of what happens to the traffic once it's on the network. Traffic shaping is a QoS technique.

6. Which flow control method resends a lost frame along with all frames sent with it?
   Answer: c. Go-Back-N sliding window
   Explanation: With the **Go-Back-N sliding window** flow control method, the sender retransmits all frames in the window, even if only one frame was lost. With the Stop-and-Wait method, only one frame is sent at a time. With the Selective Repeat sliding window method, only the lost frame is resent while successive frames continue to be sent. Backpressure is a congestion control technique.

7. Which of the following statements is true?
   Answer: c. When streaming a movie, the transmission is sensitive to delays and tolerant of loss.
   Explanation: Delayed traffic for a streaming movie results in reduced quality, while an occasional skipped video frame will likely not be noticeable. This means that, **when streaming a movie, the transmission is sensitive to delays and tolerant of loss**. An email, however, can tolerate a brief delay, but lost data can result in a corrupted message. Therefore, when sending an email, the transmission is sensitive to loss and tolerant of delays.

Section 12-3: Plan Response and Recovery Strategies

8. When repairing a coworker's computer, you find some illegal files. What should you do next?
   Answer: c. Disconnect the computer from the network and leave it running.
   Explanation: First secure the area by **disconnecting the computer from the network** (remove the Ethernet cable or disable the Wi-Fi antenna). Ideally, you should **leave the device running** without closing any applications or files. Don't shut down the computer unless a destructive program is running, and then you would immediately unplug the computer. Don't change anything on the computer by taking screenshots, and don't tamper with evidence by deleting files.

9. Which backup site includes a running server that does not necessarily have access to the latest backups?
Answer: a. Warm site
Explanation: At a **warm site**, you might have server duplicates configured, updated, and connected, but backed up data is applied only at regular intervals to reduce costs. At a cold site, devices aren't necessarily configured or connected. At a hot site, servers are constantly mirrored so they have access to the latest data. On site refers to the location of resources.

10. Which power device prevents a critical server from losing power, even for an instant?
Answer: d. UPS
Explanation: Because the server never needs to switch from the wall outlet's power to the **UPS's (uninterruptible power supply)** power, there is no risk of briefly losing service. A surge protector redirects excess voltage away from the device to a ground, thereby protecting the device from harm. A generator serves as a backup power source for many devices, providing power redundancy in the event of a total blackout. A PDU (power distribution unit) acts as a power strip to bring power from outlets, a generator, or a UPS closer to the devices on the rack.